

char buf[1024];
s.getChar(buf, end, buf, 0)

1) Student Management System

Pseudo code:

class student

{

Private name, rollNO, marks1, marks2, marks3, average

Method acceptDetails()

{

Read name, roll no, mark1, marks 2, mark3

}

Method calculateAverage()

{

average = (marks1 + marks 2 + marks3) / 3

}

method displayResult()

{
 Display name, roll NO, average

 If average ≥ 50
 Display "Pass"

 Else
 Display "Fail"

}

main()

{
 create student object

 acceptDetails()

 calculateAverage()

 displayResult()

}

}

2) Library Book Management

Pseudo code:-

class Book

{
 Private title, author, ISBN,

 is Available

 Method issueBook()

{
 isAvailable = false

}

 Method returnBook()

{

 isAvailable = true

}

 Method displayBook()

{

 Display title, author, ISBN, isAvailable.

```
}  
Main ()
```

```
{  
    create Book 1  
    create Book 2  
    create Book 3  
  
    Issue Book 1  
    Return Book 2  
    Display all Books  
}
```

3) Employee salary calculation:-

Pseudo code:-

```
class Employee
```

```
{  
    Private empId, name, basicPay, designation.
```

```
    Private grossSalary, netSalary  
    method calculateGrossSalary()
```

```
{  
    HRA = basicPay  $\times$  20%.
```

```
    DA = basicPay  $\times$  10%.
```

```
    grossSalary = basicPay + HRA + DA
```

```
}  
method calculateNetSalary()
```

```
{  
    PF = basicPay  $\times$  5%.
```

```
    Tax = basicPay  $\times$  10%.
```

```
    netSalary = grossSalary - PF - Tax
```

```
}  
method displayPaySlip()
```

```
{
```

Display emp
Display name
Display designation
Display grossSalary
Display netSalary

}
main()

{
create Employee object
calculate Gross salary()
calculate Net salary()
display Pay Slip()
}

}

4) Shape Area calculator using Polymorphism:-

Pseudo code:-

class Shape

{
method calculateArea()
}

class Circle extends Shape

{
method calculateArea()
}

{
area = $3.14 \times \text{radius} \times \text{radius}$

Display area
}

class Rectangle extends Shape

{
method calculateArea()
}

{
area = length $\times$ breadth

Display area
}

class Triangle extends Shape

```

    {
        Method calculateArea()
    {
        Area = 0.5 x base x height
        Display area
    }
}
main()
{
    create circle object
    create Rectangle object
    create triangle object
    calculateArea()
}

```

5) Vehicle Registration system

Pseudo code:-

```

Start
Input vehicle details
If Two Wheeler
    Display Engine capacity
Else
    Display seating capacity, AC
status
    Display vehicle details
End.

```

6) Bank Account Hierarchy

Pseudo code:-

```

class BankAccount
    deposit()
    withdraw()

class SavingsAccount extends
    BankAccount
    calculateInterest()

```

withdrawal()
class CurrentAccount extends
BankAccount
withdrawal()

Main
create SavingsAccount
deposit()
withdrawal()
calculateInterest()
create CurrentAccount
deposit()
withdrawal()

7) Hospital Patient Record System

Pseudo code:-

class Patient
generate Bill()

class InPatient extends Patient
generate Bill()

class OutPatient extends Patient
generate Bill()

Main

create Inpatient
generate Bill()

create Out Patient
generate Bill()

8) E-Commerce Product Catalog

class Product
applyDiscount()

class Electronics extends Product
applyDiscount()

class clothing extends Product

main
apply Discount()
create Electronics
apply Discount()
create clothing
apply Discount()
create Grocery
apply Discount()

9) University staff Hierarchy:-

Pseudocode:-

```
class staff  
    calculate Allowance()  
class Teachingstaff extends staff  
class Professor extends Teachingstaff  
    calculate Allowance()  
class Lecturer extends Teachingstaff  
    calculate Allowance()  
class NonTeachingstaff extends staff  
    calculate Allowance()
```

main

```
create Professor  
calculate Allowance()  
create Lecturer  
calculate Allowance()  
create NonTeachingstaff  
calculate Allowance().
```

10) Smart Home Device Controller:-

Pseudocode:-

class Smart Device

executeCommand()

class SmartLight extends SmartDevice

executeCommand()

class smartthermostat extends

Smart Device

executeCommand()

main

create SmartLight

executeCommand()

create smartthermostat

executeCommand()

create smartlock

executeCommand()

